Supplementary Table 2.1. Results from Monte Carlo simulations comparing bias in causal parameter estimates from the four existing MR methods and the proposed method: Scenario's A and B as a function of instrument strength.

Instrument Strength (α_1)	MR Method	Scenario A		Scenario B	
		Num Converged	Median (Q_1, Q_3) Bias	Num Converged	Median (Q_1, Q_3) Bias
0.01	Wald	7991	0.12 (-0.86, 1.12)	7990	-0.65 (-1.62, 0.35)
	2SPS	7991	0.48 (-3.45, 4.48)	7990	0.38 (-3.49, 4.41)
	2SRI	8000	0.48 (-3.49, 4.52)	8000	0.45 (-3.86, 4.91)
	GMM	3809	0.17 (-1.47, 1.85)	6482	-0.09 (-3.76, 6.99)
	IV-MVB	7946	-0.03 (-4.51, 3.53)	7919	-0.82 (-6.71, 2.49)
0.1	Wald	7986	0.08 (-0.76, 0.92)	7988	-0.71 (-1.57, 0.12)
	2SPS	7986	0.33 (-3.07, 3.71)	7988	0.14 (-3.30, 3.48)
	2SRI	8000	0.34 (-3.10, 3.75)	8000	0.18 (-3.67, 3.88)
	GMM	4195	0.17 (-1.42, 1.76)	6564	-0.28 (-3.58, 3.51)
	IV-MVB	7929	-0.03 (-3.80, 3.03)	7907	-0.78 (-5.34, 2.06)
0.2	Wald	7996	0.04 (-0.58, 0.62)	7997	-0.74 (-1.33, -0.15)
	2SPS	7996	0.16 (-2.31, 2.49)	7997	0.04 (-2.33, 2.44)
	2SRI	8000	0.15 (-2.36, 2.51)	8000	0.06 (-2.58, 2.72)
	GMM	5091	0.14 (-1.26, 1.54)	6793	-0.28 (-2.49, 2.00)
	IV-MVB	7959	-0.00 (-2.61, 2.19)	7927	-0.43 (-3.30, 1.72)
0.3	Wald	7998	0.01 (-0.44, 0.42)	7999	-0.77 (-1.19, -0.38)
	2SPS	7998	0.04 (-1.78, 1.68)	7999	-0.08 (-1.75, 1.50)
	2SRI	8000	0.03 (-1.81, 1.71)	8000	-0.08 (-1.95, 1.68)
	GMM	6180	0.07 (-1.27, 1.32)	7151	-0.29 (-1.88, 1.25)
	IV-MVB	7984	-0.03 (-1.90, 1.57)	7974	-0.23 (-2.00, 1.32)
0.5	Wald	8000	0.00 (-0.25, 0.25)	8000	-0.78 (-1.02, -0.54)
	2SPS	8000	-0.00 (-1.01, 1.01)	8000	-0.08 (-1.09, 0.88)

Instrument Strength (α_1)	MR Method	Scenario A		Scenario B	
		Num Converged	Median (Q_1, Q_3) Bias	Num Converged	Median (Q_1, Q_3) Bias
	2SRI	8000	-0.00 (-1.03, 1.01)	8000	-0.10 (-1.22, 0.98)
	GMM	7523	0.02 (-0.93, 0.98)	7635	-0.15 (-1.14, 0.84)
	IV-MVB	7998	-0.00 (-1.02, 1.00)	7998	-0.09 (-1.10, 0.86)
0.7	Wald	8000	0.00 (-0.18, 0.17)	8000	-0.78 (-0.95, -0.61)
	2SPS	8000	0.01 (-0.73, 0.72)	8000	-0.09 (-0.80, 0.62)
	2SRI	8000	0.01 (-0.75, 0.72)	8000	-0.11 (-0.90, 0.67)
	GMM	7922	0.02 (-0.73, 0.72)	7907	-0.10 (-0.80, 0.64)
	IV-MVB	8000	0.01 (-0.73, 0.72)	8000	-0.09 (-0.80, 0.62)
1.0	Wald	8000	-0.00 (-0.12, 0.12)	8000	-0.78 (-0.90, -0.66)
	2SPS	8000	-0.00 (-0.53, 0.51)	8000	-0.08 (-0.59, 0.43)
	2SRI	8000	-0.01 (-0.54, 0.51)	8000	-0.11 (-0.68, 0.45)
	GMM	7997	-0.00 (-0.53, 0.51)	7996	-0.08 (-0.59, 0.45)
	IV-MVB	8000	0.00 (-0.53, 0.51)	8000	-0.08 (-0.60, 0.43)

Supplementary Table 2.2. Results from Monte Carlo simulations comparing bias in causal parameter estimates from the four existing MR methods and the proposed method: Scenario's C and D across various sample sizes.

Sample Size	MR Method	Scenario C		Scenario D	
		Num Converged	Median (Q_1, Q_3) Bias	Num Converged	Median (Q_1, Q_3) Bias
50	Wald	7878	0.03 (-0.64, 0.77)	7890	-0.76 (-1.36, 0.00)
	2SPS	7878	0.23 (-2.87, 3.34)	7890	0.15 (-2.65, 3.39)
	2SRI	8000	0.25 (-2.89, 3.36)	8000	0.13 (-2.99, 3.70)
	GMM	6372	0.10 (-2.37, 2.73)	6697	-0.21 (-2.67, 2.89)
	IV-MVB	7384	0.05 (-3.17, 3.13)	6793	-0.23 (-3.31, 2.46)
200	Wald	7992	0.00 (-0.40, 0.38)	7994	-0.77 (-1.17, -0.38)
	2SPS	7992	0.02 (-1.65, 1.63)	7994	-0.00 (-1.72, 1.62)
	2SRI	8000	0.01 (-1.69, 1.64)	8000	-0.03 (-1.94, 1.76)
	GMM	6705	0.03 (-1.37, 1.35)	7018	-0.27 (-1.88, 1.17)
	IV-MVB	7982	-0.00 (-1.73, 1.57)	7960	-0.13 (-1.88, 1.45)
500	Wald	8000	0.00 (-0.25, 0.25)	8000	-0.78 (-1.03, -0.54)
	2SPS	8000	0.02 (-1.02, 1.04)	8000	-0.09 (-1.14, 0.94)
	2SRI	8000	0.01 (-1.04, 1.05)	8000	-0.12 (-1.28, 1.02)
	GMM	7510	0.03 (-0.93, 1.01)	7606	-0.17 (-1.20, 0.86)
	IV-MVB	7999	0.01 (-1.03, 1.03)	7998	-0.10 (-1.15, 0.92)
1000	Wald	8000	-0.00 (-0.18, 0.17)	8000	-0.78 (-0.96, -0.61)
	2SPS	8000	-0.01 (-0.75, 0.69)	8000	-0.10 (-0.82, 0.62)
	2SRI	8000	-0.02 (-0.76, 0.69)	8000	-0.12 (-0.92, 0.67)
	GMM	7924	-0.01 (-0.74, 0.69)	7896	-0.11 (-0.83, 0.63)
	IV-MVB	8000	-0.02 (-0.75, 0.68)	8000	-0.10 (-0.82, 0.61)
1500	Wald	8000	-0.00 (-0.14, 0.14)	8000	-0.78 (-0.92, -0.64)
	2SPS	8000	-0.01 (-0.59, 0.58)	8000	-0.08 (-0.68, 0.48)
	2SRI	8000	-0.01 (-0.61, 0.59)	8000	-0.11 (-0.77, 0.51)

Sample Size	MR Method	Scenario C		Scenario D	
		Num Converged	Median (Q_1, Q_3) Bias	Num Converged	Median (Q_1, Q_3) Bias
	GMM	7985	-0.01 (-0.60, 0.59)	7978	-0.08 (-0.69, 0.51)
	IV-MVB	8000	-0.01 (-0.59, 0.58)	8000	-0.09 (-0.69, 0.48)
2000	Wald	8000	-0.00 (-0.13, 0.12)	8000	-0.78 (-0.90, -0.66)
	2SPS	8000	-0.01 (-0.52, 0.50)	8000	-0.07 (-0.58, 0.42)
	2SRI	8000	-0.01 (-0.53, 0.50)	8000	-0.09 (-0.65, 0.45)
	GMM	7999	-0.01 (-0.52, 0.50)	7998	-0.07 (-0.58, 0.45)
	IV-MVB	8000	-0.00 (-0.52, 0.50)	8000	-0.08 (-0.58, 0.41)
2500	Wald	8000	0.00 (-0.11, 0.11)	8000	-0.78 (-0.89, -0.67)
	2SPS	8000	0.01 (-0.46, 0.46)	8000	-0.08 (-0.54, 0.35)
	2SRI	8000	0.01 (-0.46, 0.46)	8000	-0.11 (-0.61, 0.38)
	GMM	7997	0.01 (-0.46, 0.47)	7997	-0.08 (-0.54, 0.38)
	IV-MVB	8000	0.01 (-0.45, 0.46)	8000	-0.09 (-0.54, 0.35)